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Fluorescent Dyes

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Diaminodicyanoquinones: Fluorescent Dyes with High Dipole Moments and Electron-Acceptor Properties

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Dedicated to Professor Hans-Ulrich Reißig on the occasion of his 70th birthday

Abstract: *Fluorescent dyes are applied in various fields of research, including solar cells and light-emitting devices, and as reporters for assays and bioimaging studies. Fluorescent dyes with an added high dipole moment pave the way to nonlinear optics and polarity sensitivity. Redox activity makes it possible to switch the molecule's photophysical properties. Diaminodicyanoquinone derivatives possess high dipole moments, yet only low fluorescence quantum yields, and have therefore been neglected as fluorescent dyes. Here we investigate the fluorescence properties of diaminodicyanoquinones using a combined theoretical and experimental approach and derive molecules with a fluorescence quantum yield exceeding 90 %. The diaminodicyanoquinone core moiety provides chemical versatility and can be integrated into novel molecular architectures with unique photophysical features.*

Applications of fluorescent dyes range from light harvesting in solar cells,[1] in light-emitting devices,[2] as reporters for the life sciences,[3] as molecular switches,[4] and in nonlinear optics.[4–5] 7,7,8,8-Tetracyanoquinodimethane (TCNQ) is a strong electron acceptor used in conductive donor–acceptor systems like TCNQ–tetrathiafulvalene compounds.[6] An interesting subclass are diamino-substituted TCNQ derivatives, namely diaminodicyanoquinones (DADQs), which possess electron-donating and -accepting moieties linked by a π-system, resulting in high dipole moments of 10–20 Debye.[7] DADQ derivatives have been discussed as novel materials, for example, for applications in nonlinear optics.[7a] However, their fluorescence quantum yields (QYs) in solution are generally below 0.5 % due to nonradiative relaxations of the excited state on the ps timescale.[8] Only in the solid state are DADQs moderately emissive with QYs of up to 45 %.[9] Consequently, DADQ derivatives are not considered as fluorescent dyes. Conformational relaxations can have a high impact on fluorescence,[10] as exploited, for example, for aggregation-induced emission,[11] where blocking these pathways is used to generate a signal amplification or used as sensing principle for viscosity[12] and temperature-responsive molecules.[10b,13]

Here we demonstrate that QYs in DADQs can be finetuned by adjusting the substitution pattern, such that QYs exceeding 90 % is possible (see Figure 1). Theoretical calculations reveal that benzene-functionalized DADQs show high QYs as their molecular structures remain close to the ground state conformation after photoexcitation and are not submitted to internal conversion (IC) or intersystem crossing (ISC). Accordingly, DADQ derivatives may now be considered as fluorescent dyes and thus applications, such as optical switches or nonlinear optics, are within reach.

First TCNQ was reacted with pyrrolidine to activate the substitution of the geminal cyano groups of TCNQ in the 7,7 position.[14] This 7-pyrrolidino-7,8,8-tricyanoquinomethane (PTCNQ) reacts with the respective amine to give compounds **1**–**4** (Figure 1 A) in 40–80 % yields. The compounds were characterized by ^{1}H and ^{13}C NMR spectroscopy, FTIR spectroscopy, elemental analysis (EA), and mass spectrometry (MS). In addition, the absorption and fluorescence properties were investigated in selected solvents.

The molar absorption coefficients are solvent dependent and reach values of up to 58000 $L\,mol^{-1}\,cm^{-1}$ in acetonitrile (ACN) as depicted in Figure 2 for **4**. While the unstructured absorption band lies between 300 and 400 nm, the fluorescence is shifted bathochromically by 40–90 nm (25×10^{4}–$11.1\times10^{4}\,cm^{-1}$).

The difference densities between the ground state and the excited state (inset Figure 2) illustrate the shift in electron density from the dicyanomethane group to the amine groups upon photoexcitation. This transition leads to a notable reduction of the dipole moment in the excited state from approximately 30 D to 22 D (Table S9). This transition is well known for DADQs, and is the reason for the observed negative solvatochromism (Figure S2 and Table S2).[15] In addition, the S_0–S_1 transition in **4** is extended across the

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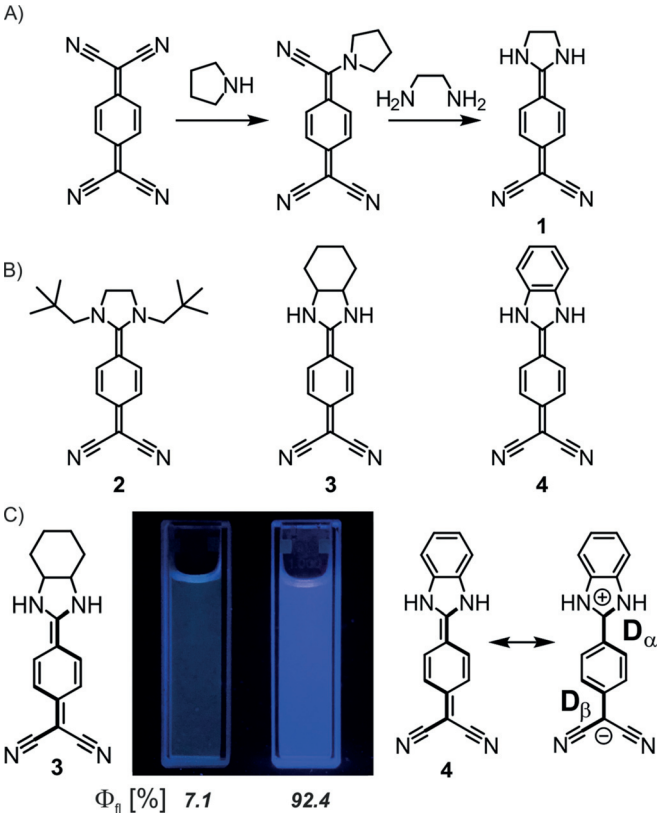


Figure 1. A) Synthetic route starting with the activation of TCNQ by reaction with pyrrolidine and subsequent reaction with the respective diamine (ethylene diamine is shown) to yield compounds **1–7.** B) Chemical structures of **1–4**. C) Photographs of **3** (left) and **4** (right) under illumination with 366 nm light with the respective fluorescence quantum yields in DMSO. Right side: Mesomeric forms of **4** and the definition of the dihedral angles D_α and D_β.

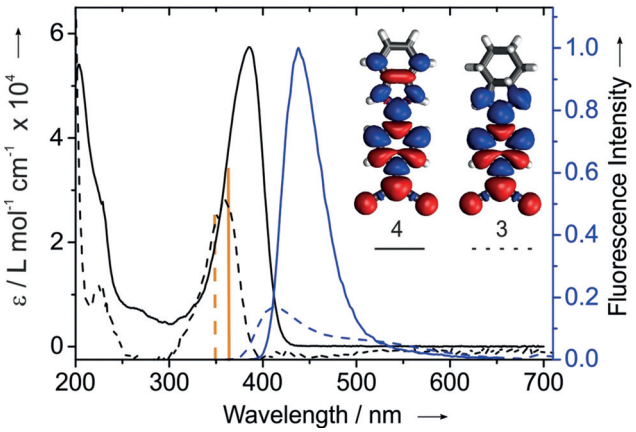


Figure 2. Absorption (black) and fluorescence spectra (blue) of **3** (dashed) and **4** (solid) in acetonitrile (normalized with respect to the fluorescence of **4**). The vertical orange bars represent transitions of compound **3** (349 nm, oscillator strength: 1.02 a.u.) and **4** (363 nm, oscillator strength: 1.30 a.u.) calculated using TD-DFT at the CAM-B3LYP level. Insets: Difference densities between ground state and excited state of **4** (left) and **3** (right) to visualize the electron flow during the electronic excitation. Blue and red areas correspond to areas of electron enhancement and depletion, respectively. Isovalue = 0.005 Å^{-3}.

enlarged π-system, including the phenyl ring. The absolute QY values are summarized in Table 1. Compound **4** shows very high QYs in all solvents, with values of 92% in dimethyl sulfoxide (Figure 1C), around 70% in tetrahydrofuran, dimethylformamide, and acetonitrile, and 56% and 35% in ethanol and methanol, respectively. In contrast, **1** and **3** show

Table 1: Fluorescence quantum yields of compounds **1–7** in solvents of different polarity. Values for the normalized Dimroth–Reichardt parameter E_T^N were taken from Ref. [16].

E_T^N	Solvent	**1** Φ [%]	**2** Φ [%]	**3** Φ [%]	**4** Φ [%]	**5** Φ [%]	**6** Φ [%]	**7** Φ [%]
0.207	THF	7	<1	7	73	34	69	<1
0.386	DMF	7	<1	10	72	62	81	3
0.444	DMSO	10	<1	7	92	39	90	<1
0.460	ACN	7	<1	12	71	18	53	<1
0.654	EtOH	<1	<1	<1	56	63	70	<1
0.762	MeOH	<1	<1	<1	35	20	30	<1

QYs in the range of 0–12% and **2** QYs of <1%. The fluorescence lifetimes of **1–4** and the calculated radiative and nonradiative rate constants (Tables S3 and S4) do not show a straightforward correlation with the QY data. QY measurements in solvents of varying viscosity confirm the assumption that rotations around the dihedral angles D_α and D_β play a significant role for the nonradiative excited-state deactivation of **1–3**. By increasing the fraction of polyethylene glycol (PEG) in ethanol (PEG–EtOH) from 0% to 75%, we observe a 3–7-fold increase in fluorescence intensity for **1–3** (Figure 3 and Figure S7), resulting in an increase of QY of **2** from 0.4% to 2.9%. In contrast, the fluorescence of **4** is barely affected by an increase in solvent viscosity (Figure S7D).

For compounds **1–4** the fluorescence intensity decreases by more than 30% with increasing temperature in the temperature range of 40°C to 110°C (Figure S9). Upon decreasing the temperature (273–173 K), the fluorescence

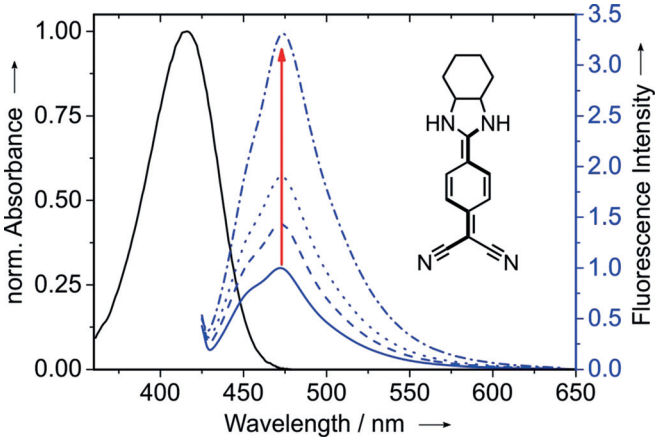


Figure 3. Absorption spectrum and emission spectra of **3** in EtOH–PEG mixtures ranging from 0 to 75% PEG in steps of 25%, depicted by the red arrow. Solid line 0%, dashed line 25%, dotted line 50%, dashed–dotted line 75%. With an increasing ratio of PEG and hence increasing viscosity, the fluorescence intensity increases by a factor of 3.3.

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intensity increases qualitatively for **1**–**4** (Figure S11). A closer look at the shape of the fluorescence signals indicates that various fluorescent states are populated (Figure S12). As we did not thoroughly study the viscosity effects and aggregation upon these temperature changes, the observed trends are qualitative.

The experimental results indicate that rotations play a major role in controlling the QY. In particular, the N-substituted derivatives **2** and **7** (Figures 1 and 4), twisted in their ground-state structure at D_α (Figure S17C), are non-fluorescent. In analogy to **4**, compounds **5** (Br derivative) and **6** (CN derivative), both with a benzene unit, are highly fluorescent (Figure 4 and Table 1).

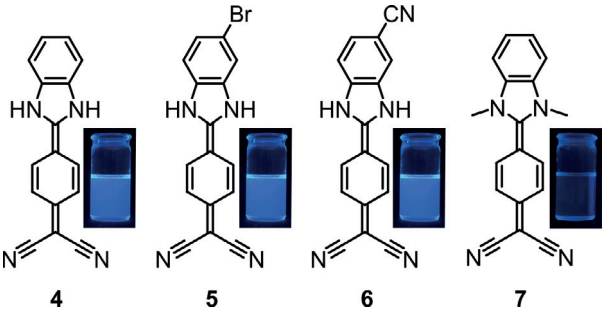


Figure 4. Chemical structures of compounds **4**–**7** with photographs showing their fluorescence (excitation wavelength 366 nm).

To gain more insight into the optical properties of our DADQ derivatives and to elucidate the remarkable QY of **4**, we performed time-dependent density functional theory (TD-DFT) calculations at the CAM-B3LYP[17] level for **1**–**4** and **7** using ACN as the solvent with CPCM ($\varepsilon = 37.5$).[18] A prerequisite for a high QY is a close similarity between the relaxed structures of the excited state and the ground state without IC or ISC events influencing the photoexcitation process.

The fluorescence deactivation of the DADQ species may be controlled by two photoinduced intramolecular torsions at dihedral angles D_α and D_β (Figure 5). To our surprise, the ground (S_0) and excited state (S_1) structures of all compounds hardly differ (Figure S16) and the oscillator strengths remain rather constant (Table S8). As this would imply all compounds show similar QYs, a more detailed look at the potential energy surfaces (PES) of the intramolecular rotations is needed. To account for the influence of multireference states we opted for the DFT/MRCI method originally proposed by Grimme and Waletzke[19] and revised by Marian et al.[20] with a Kohn–Sham orbital basis generated at the CPCM/BHLYP/def2-TZVP[21] level to obtain accurate energies. Test calculations showed that the rotational barriers at D_α and D_β do not critically influence each other (Figure S17A,B), which is why we will treat D_α and D_β separately. In the ground state S_0, rotation around D_β at the Franck–Condon point is hindered by large barriers of around 60 kJ mol^{-1} for all compounds, but a rotation around D_α is possible except for **2** and **7** where the N substituents block the way (Figures S17C and S18A). In contrast, in the excited

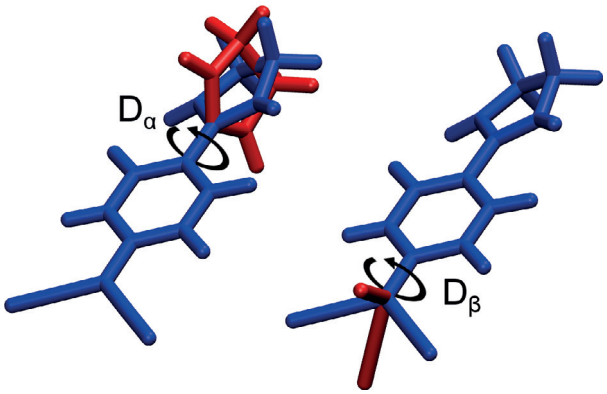


Figure 5. Two kinds of torsion angles (shown for **1**) that affect the fluorescence deactivation mechanism in DADQs: D_α (left) and D_β (right).

state S_1, rotation around D_α is prevented with barriers of 60–100 kJ mol^{-1}, but rotation around D_β shows barriers of around 12 kJ mol^{-1} (**1**–**3**) to 25 kJ mol^{-1} (**4**, **7**) (Figures S17D and S18B). Large barriers arise as a rotation around D_β diminishes the basis function overlap between the π-orbitals centered at the phenyl ring and those at the nitrile groups in the HOMO, while a rotation around D_α decreases the overlap between the phenyl ring's π-orbitals and the p-orbitals of the amino groups in the LUMO. Low barriers are explained by nodes present in the HOMO at D_α and in the LUMO at D_β, where an intramolecular torsion would not impede any orbital overlap (Figure 6).

The PES at D_α is less important for **2**, **3**, and **4**, where an electronic excitation starting from S_0 on the S_1 PES will just recover the S_0 minimum structure. However, this torsion may account for fluorescence quenching by IC or ISC in **7** and slightly in **1**. For **1**–**4**, QY loss is expected to mainly occur due to the relatively small rotational barrier in the S_1 at D_β (Figure 7). As expected, the respective rotational barrier is highest for **4** and lowest for **1**. This can be attributed to the stabilization of the ground state geometry of **4** through the extended π-system. Using the Arrhenius' equation at $T =$

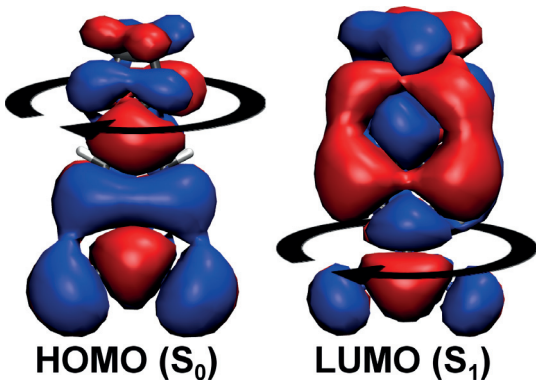


Figure 6. HOMO (left) and LUMO (right) obtained at the BHLYP/def2-TZVP level. Rotations (indicated by arrows) are favored due to nodal planes in the orbitals, while orbital overlap keeps the structures rigid at D_β in the HOMO (left) and at D_α in the LUMO (right), isovalue = 0.003 Å^{-3}.

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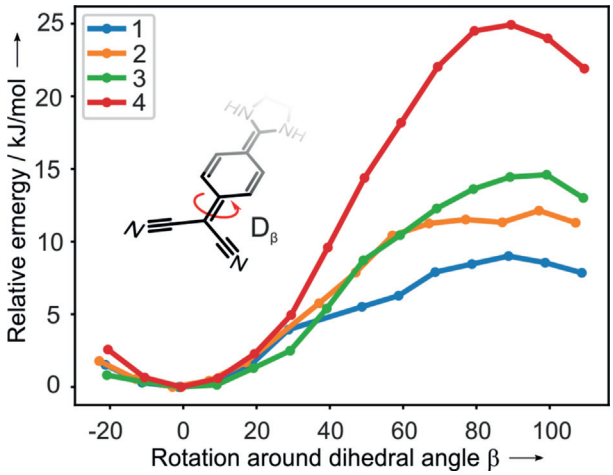


Figure 7. PES scan of the S_1 state around dihedral angle D_β for **1**–**4** computed at the CPCM/BHLYP/def2-TZVP/MRCI level. **7** is omitted as its fluorescence deactivation mechanism is different.

300 K, we define k_{rot}^{-1} (the reciprocal rate constant for the rotation around D_β; see Equation S5) as a relative measure to compare rotational barriers. Relating k_{rot}^{-1} for **1**–**4** illustrates that **4** is up to 480 times more likely than **1**–**3** to fluoresce after photoexcitation into the S_1.

$$k_{rot}^{-1}(\mathbf{1}) : k_{rot}^{-1}(\mathbf{2}) : k_{rot}^{-1}(\mathbf{3}) : k_{rot}^{-1}(\mathbf{4}) = 1 : 3 : 12 : 481$$

ISC events between singlet and triplet surfaces contribute to fluorescence quenching in all compounds. For **7** the most likely deactivation mechanism is via an ISC of the S_1 with T_1 and T_2 along D_α (Figure S19 A). This mechanism may play a role for **1** as well; however, it is assumed to have a smaller impact (Figure S19 B). At D_β all compounds show a singlet/triplet separation energy of around 0.7 eV at the Franck–Condon point, which should be just large enough to prevent ISC events. However, as the S_1 and the T_0 eventually come very close at an angle of around 90° (Figure S20) in all molecules, the rotational barrier presented by k_{rot}^{-1} and the nonradiative relaxation through ISC events are directly linked. Since **1**–**3** are much more likely to overcome the rotational barrier at D_β than **4,** their QYs are significantly lower.

While this qualitatively accounts for the drastic differences in QY, it does not fully explain the large QY of **4** by itself. As solvent effects have only been included in an implicit way, an explicit solvent environment of ACN was set up (Figure S21 A) and the rotational barrier at D_β was recalculated for **1** and **4** with the smaller def2-SVP basis set and fixed solvent molecules to save computational resources (Figure S21 B). While the total barrier hardly changed for **1**, the barrier height for **4** increased by roughly 30 %. Thus, the solvent shell may play a significant role at least in the case of **4**, where it likely contributes to its large QY.

To further characterize **1**–**4**, we performed cyclic voltammetry (CV) experiments in a 0.1 M Bu_4NPF_6 solution of DMF (Figure S14). The molecules show a complex redox behavior, with mostly reversible oxidations (for **4**) and irreversible reductions (quasi reversible for **2**), which likely involve the proton of the secondary amine and provide a hint for an electron transfer followed by a chemical reaction (EC mechanism).[22] As expected, the replacement of the cyano moieties with secondary and tertiary amines favors oxidation processes, which are not observed for TCNQ (Figure S15). This also renders the molecules more electron-rich and causes a cathodic shift of the reduction potentials. (see Table S7 for more details). The HOMO–LUMO gaps were calculated at the CPCM/BHLYP/MRCI level. The trends between experiment and theory show overall agreement (Table 2). The electrochemically measured gaps differ slightly in absolute values, as the solvent model neglects bulk effects at the electrode.[23]

Table 2: Electrochemical, theoretical, and optical data of compounds **1**–**4** with the resulting orbital energies and HOMO–LUMO gaps (E_g) [eV].

	CV data[a]			Optical data[b]	Theory[c]
	E_{HOMO} [eV]	E_{LUMO} [eV]	E_g^{el} [eV]	E_g^{opt} [eV]	E_g^{theo} [eV]
1	−2.37	0.16	2.53	3.31	3.24
2	−2.42	0.12	2.54	2.87	3.49
3	−2.27	−0.14	2.13	3.16	3.17
4	−2.22	−0.21	2.01	2.94	2.98

[a] Orbital energies calculated from the first reduction or oxidation half-wave potentials referenced to FcH/FcH^+(−4.80 eV): $\varepsilon = (-(E_{1/2}-4.80)$ eV [b] The optical HOMO–LUMO gap was determined from the onset at the red edge of the absorption band [c] CPCM/BHLYP/def2-TZVP/MRCI.

In conclusion, we synthesized highly fluorescent diaminodicyanoquinones showing quantum yields of over 90 % when benzene-substituted. We attribute the high fluorescence quantum yields mainly to restricted internal rotations, while quenching occurs through the intersection of singlet and triplet states. However, other effects, such as molecular aggregation, may contribute as well. The electrochemical characterization of these molecules reveals complex redox activity. In summary, we could show that diaminodicyanoquinone derivatives are a novel class of fluorescent dyes customizable for photophysical applications, the potential of which needs to be studied in more detail.

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Conflict of interest

The authors declare no conflict of interest.

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